Set-Coded Turing Machines in Isabelle/ZF

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1 Research question

An EPQ investigation of the continuum hypothesis and the computability of model invariance motivated this development. The formalisation requires a precise connection between external computation and truth inside set-theoretic models. The library defines deterministic Turing machines as sets, their finite execution, numerical machine encodings, and tape equivalence that ignores surplus blank cells.

2 Verified connection

For each well-formed machine P and finite symbol input x, the construction halt_sentence(P,x) is an explicit first-order sentence with no free variables. For every transitive ZF set model M, the following equivalence is proved.

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M satisfies halt_sentence(P,x) iff P halts on x.
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The sentence depends only on P and x. A finite halting trace provides an internal witness, and an internal witness determines the corresponding external finite execution. Model existence is not asserted.

3 Effective sentence construction

For independently supplied natural numbers e and n, a primitive-recursive constructor returns the exact code of halt_sentence(decode_machine(e),numeral_input(n)). Both membership in the primitive-recursive class and equality to this formula code are proved. The self-input case takes n equal to e. The certificate concerns numerical construction.

4 Use, verification and scope

The standalone AFP candidate contains 17 theories. Client examples derive program properties, transfer them across equivalent tapes, and obtain model-satisfaction statements through named interfaces. The recorded GitHub Actions build checks the candidate and separate research sessions with Isabelle2025-2 and the AFP snapshot of 6 October 2026. Research extensions remain separate, and the full EPQ argument is not yet formalised. The candidate has not been submitted to or accepted by AFP.

Repository | Main theorem | Recorded build | EPQ paper
github.com/anguuananguuan-droid/set-coded-computation-zf